



INTERNATIONAL ISLAMIC UNIVERSITY, ISLAMABAD

SOFTWARE ENGINEERING DEPARTMENT

Total Marks: 20

Obtained Marks: _____

Final Semester Project Report

Last date of Submission: 3 June 2026

Submitted To: Shakeel Ahmad

**Student Names: Muhammad Ali Asghar
Saad Saleem
Ali Abbas**

**Reg. Numbers: 004385 / BSSE / F24
004349 / BSSE / F24
004364 / BSSE / F24**



Final Semester Project Report

Subject: Database Systems

Project Title:

Fitness and Health Centre using Oracle 10g Express Edition

Introduction

- A Gym Management System is designed to maintain and organize gym related records efficiently through a relational database.
- Managing gym operations manually can become difficult because many records are involved such as members, trainers, workout sessions, attendance, invoices, equipment and feedback.
- The purpose of this project is to design and implement a relational database system using Oracle 10g Express Edition.
- This system stores and manages:
 1. Gym members
 2. Membership plans
 3. Trainers
 4. Nutritionists
 5. Workout sessions
 6. Equipment
 7. Attendance
 8. Billing and invoices
 9. Diet plans
 10. Member feedback
- The project applies SQL concepts including:
 1. Table creation
 2. Primary and foreign keys
 3. Constraints
 4. Data insertion
 5. Joins
 6. Group By
 7. Having clause
 8. Subqueries
 9. Views
 10. Triggers



Objectives of the Project

- The main objectives are:
 1. To design a relational database for gym management
 2. To create multiple related tables
 3. To apply primary key and foreign key relationships
 4. To insert sample records
 5. To perform business queries
 6. To create SQL views
 7. To implement triggers
 8. To manage gym data efficiently

Tools and Technologies Used

- **Database Platform**
 1. Oracle 10g Express Edition
- **Query Tool**
 2. Oracle SQL Command Line
- **Language**
 3. SQL

Database Concepts Used

- CREATE TABLE
- INSERT INTO
- SELECT
- JOIN
- GROUP BY
- HAVING
- VIEW
- TRIGGER
- CONSTRAINTS



The following tables were created.

- Stores basic information of all persons.
- **Attributes:**
 1. PERSON_ID
 2. NAME
- **Screenshot:**

2) PERSON_PHONE

- Stores phone numbers.
- **Attributes:**
 1. PERSON_ID
 2. PHONE
- **Screenshot:**

[illegible]



- Stores regular member features.
- **Screenshot:**

- Stores premium member details.
- **Screenshot:**

- Stores trainer data.
- **Screenshot:**

Results	Explain	Describe	Saved SQL	History
-------------------------	-------------------------	--------------------------	---------------------------	-------------------------

Object Type	TABLE	Object	TRAINER
-------------	-------	--------	---------

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
TRAINER	TRAINER_ID	Number	-	-	-	1	-	-	-
	PERSON_ID	Number	-	-	-	-	✓	-	-
	SPECIALTY	Varchar2	100	-	-	-	✓	-	-

1 - 3



- Stores nutritionist data.
- **Screenshot:**

9) WORKOUT_SESSION

- Stores workout schedule.
- **Screenshot:**

10) EQUIPMENT

- Stores equipment inventory.
- **Screenshot:**

Results	Explain	Describe	Saved SQL	History
-------------------------	-------------------------	--------------------------	---------------------------	-------------------------

Object Type	TABLE	Object	EQUIPMENT
-------------	-------	--------	-----------

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EQUIPMENT	EQUIPMENT_ID	Number	-	-	-	1	-	-	-
	NAME	Varchar2	100	-	-	-	-	-	-
	TYPE	Varchar2	50	-	-	-	✓	-	-
	QUANTITY	Number	-	-	-	-	✓	0	-

1 - 4



- Stores equipment assigned to sessions.
- **Screenshot:**

- Stores attendance.
- **Screenshot:**

- Stores trainer assignments.
- **Screenshot:**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
ASSIGNED_TRAINER	MEMBER_ID	Number	-	-	-	1	-	-	-
	TRAINER_ID	Number	-	-	-	2	-	-	-

1 - 2



- Stores billing details.
- **Screenshot:**

Results Explain Describe Saved SQL History

Object Type TABLE Object INVOICE

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
INVOICE	INVOICE_ID	Number	-	-	-	1	-	-	-
	MEMBER_ID	Number	-	-	-	-	✓	-	-
	INVOICE_DATE	Date	7	-	-	-	✓	-	-
	AMOUNT	Number	-	10	2	-	✓	-	-
1 - 4									

- Stores nutrition plans.
- **Screenshot:**

[illegible]

- Stores member comments.
- **Screenshot:**

Results	Explain	Describe	Saved SQL	History
-------------------------	-------------------------	--------------------------	---------------------------	-------------------------

Object Type **TABLE** Object **FEEDBACK**

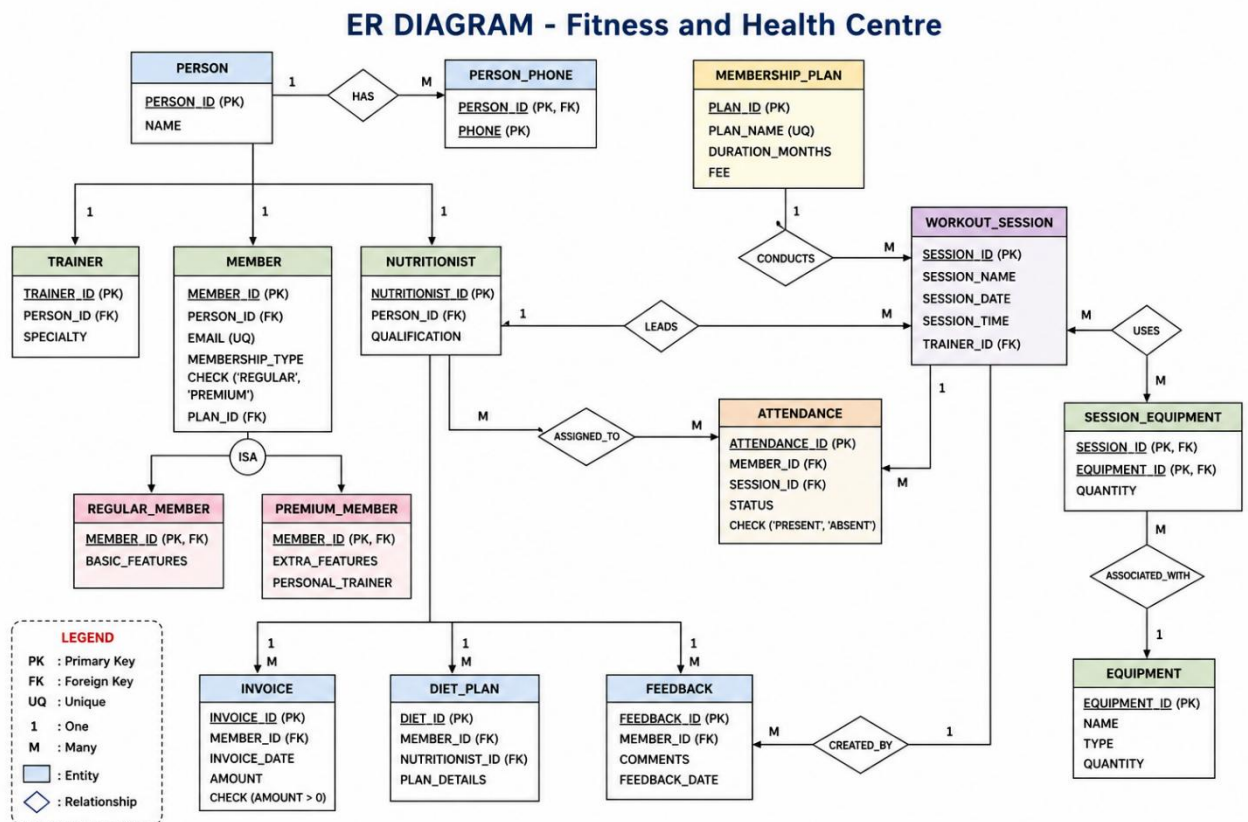
Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
FEEDBACK	FEEDBACK_ID	Number	-	-	-	1	-	-	-
	MEMBER_ID	Number	-	-	-	-	✓	-	-
	COMMENTS	Varchar2	500	-	-	-	✓	-	-
	FEEDBACK_DATE	Date	7	-	-	-	✓	-	-

1 - 4



Relationships in Database

- The system contains multiple relationships.
- Examples:**
 - PERSON → PERSON_PHONE = One to Many
 - PERSON → MEMBER = One to One
 - PERSON → TRAINER = One to One
 - PERSON → NUTRITIONIST = One to One
 - MEMBER → INVOICE = One to Many
 - MEMBER → FEEDBACK = One to Many
 - MEMBER ↔ TRAINER = Many to Many
 - WORKOUT_SESSION ↔ EQUIPMENT = Many to Many
- ERD Screenshot:**





Data Population

- Sample records were inserted successfully.
- **Inserted records include:**

1) Persons

- **Screenshot:**

Results	Explain	Describe
PERSON_ID		NAME
1		Ahmed Khan
2		Ali Raza
3		Usman Tariq
4		Hassan Shah
5		Bilal Ahmed
6		Saad Malik
7		Hamza Ali
8		Talha Khan
9		Zain Abbas
10		Omer Farooq
11		Ayesha Noor
12		Fatima Khan
13		Sara Ahmed
14		Maryam Ali
15		Hina Tariq

2) Phone Numbers

- **Screenshot:**

Results	Explain	Describe
PERSON_ID		PHONE
1		03001111111
2		03002222222
3		03003333333
4		03004444444
5		03005555555
6		03006666666
7		03007777777
8		03008888888
9		03009999999
10		03101111111
11		03102222222
12		03103333333
13		03104444444
14		03105555555
15		03106666666



3) Membership Plans

- Screenshot:

Results	Explain	Describe	Saved SQL	History
PLAN_ID	PLAN_NAME	DURATION_MONTHS	FEE	
1	Basic	1	3000	
2	Silver	3	8000	
3	Gold	6	15000	
4	Premium	12	25000	
5	VIP	12	40000	

4) Trainers

- Screenshot:

Results	Explain	Describe	Saved SQL	History
TRAINER_ID	PERSON_ID	SPECIALTY		
1	1	Weight Training		
2	2	Cardio		
3	3	Body Building		
4	4	CrossFit		
5	5	Yoga		

5) Nutritionists

- Screenshot:

Results	Explain	Describe	Saved SQL	History
NUTRITIONIST_ID	PERSON_ID	QUALIFICATION		
1	11	BS Nutrition		
2	12	MS Dietetics		
3	13	Clinical Nutrition		



6) Members

○ Screenshot:

Results	Explain	Describe	Saved SQL	History
MEMBER_ID	PERSON_ID	EMAIL	MEMBERSHIP_TYPE	PLAN_ID
1	6	saad@gmail.com	REGULAR	1
2	7	hamza@gmail.com	PREMIUM	4
3	8	talha@gmail.com	REGULAR	2
4	9	zain@gmail.com	PREMIUM	5
5	10	omer@gmail.com	REGULAR	3

7) Workout sessions

○ Screenshot:

Results	Explain	Describe	Saved SQL	History
SESSION_ID	SESSION_NAME	SESSION_DATE	SESSION_TIME	TRAINER_ID
1	Morning Cardio	01-JUN-26	08:00 AM	2
2	Strength Training	02-JUN-26	10:00 AM	1
3	CrossFit Session	03-JUN-26	06:00 PM	4

8) Equipment Records

○ Screenshot:

Results	Explain	Describe	Saved SQL	History
EQUIPMENT_ID	NAME	TYPE	QUANTITY	
1	Treadmill	Cardio	5	
2	Dumbbells	Strength	20	
3	Bench Press	Strength	4	
4	Exercise Bike	Cardio	3	
5	Yoga Mat	Flexibility	15	

9) Attendance Records

○ Screenshot:

Results	Explain	Describe	Saved SQL	History
ATTENDANCE_ID	MEMBER_ID	SESSION_ID	STATUS	
1	1	1	PRESENT	
2	2	1	ABSENT	
3	3	2	PRESENT	
4	4	2	PRESENT	
5	5	3	ABSENT	



10) Invoices

- Screenshot:

Results	Explain	Describe	Saved SQL	History
INVOICE_ID	MEMBER_ID	INVOICE_DATE	AMOUNT	
1	1	01-JUN-26	3000	
2	2	01-JUN-26	25000	
3	3	02-JUN-26	8000	
4	4	02-JUN-26	40000	
5	5	03-JUN-26	15000	

11) Diet Plans

- Screenshot:

Results	Explain	Describe	Saved SQL	History
DIET_ID	NUTRITIONIST_ID	PLAN_DETAILS		
1	1	Weight Loss Diet		
2	2	Muscle Gain Diet		
3	3	Balanced Nutrition Plan		

12) Feedback

- Screenshot:

Results	Explain	Describe	Saved SQL	History
FEEDBACK_ID	MEMBER_ID	COMMENTS	FEEDBACK_DATE	
1	1	Excellent Trainer	05-JUN-26	
2	2	Very Good Session	05-JUN-26	
3	3	Equipment Needs Upgrade	06-JUN-26	



Business Queries

- Different business queries were performed.

1. Members with Membership Plans

- **Purpose:**

Display members along with plan and fee.

- **Screenshot:**

Results	Explain	Describe	Saved SQL	History
MEMBER_ID	NAME	PLAN_NAME	FEE	
1	Saad Malik	Basic	3000	
2	Hamza Ali	Premium	25000	
3	Talha Khan	Silver	8000	
4	Zain Abbas	VIP	40000	
5	Omer Farooq	Gold	15000	

2. Trainer with Workout Sessions

- **Purpose:**

Display trainer sessions.

- **Screenshot:**

Results	Explain	Describe	Saved
TRAINER_ID	SESSION_NAME		
2	Morning Cardio		
1	Strength Training		
4	CrossFit Session		



3. Attendance Details

- **Purpose:**

Display attendance status.

- **Screenshot:**

Results	Explain	Describe	Saved SQL	History
ATTENDANCE_ID	MEMBER_ID	SESSION_ID	STATUS	
1	1	1	PRESENT	
2	2	1	ABSENT	
3	3	2	PRESENT	
4	4	2	PRESENT	
5	5	3	ABSENT	

4. GROUP BY

- **Purpose:**

Count members by plan.

- **Screenshot:**

Results	Explain	Describe
PLAN_ID	COUNT(*)	
1	1	
2	1	
4	1	
5	1	
3	1	



5. HAVING Clause

- **Purpose:**

Display grouped plans.

- **Screenshot:**

Results	Expl.
no data found	

Note:

- No data found because no membership plan had more than one member, so the **HAVING COUNT(*) > 1** condition was not satisfied

6. Subquery

- **Purpose:**

Members with invoice above average.

- **Screenshot:**

Results	Explain	Describe	Saved SQL	History
MEMBER_ID	PERSON_ID	EMAIL	MEMBERSHIP_TYPE	PLAN_ID
2	7	hamza@gmail.com	PREMIUM	4
4	9	zain@gmail.com	PREMIUM	5

7. IN Query

- **Purpose:**

Members present in attendance.

- **Screenshot:**

Results	Explain	Describe	Saved SQL	History
MEMBER_ID	PERSON_ID	EMAIL	MEMBERSHIP_TYPE	PLAN_ID
1	6	saad@gmail.com	REGULAR	1
3	8	talha@gmail.com	REGULAR	2
4	9	zain@gmail.com	PREMIUM	5



8. Equipment with Session

- **Purpose:**

Show equipment usage.

- **Screenshot:**

Results	Explain	Describe	Savi
SESSION_NAME	NAME		
Morning Cardio	Treadmill		
Strength Training	Dumbbells		
Strength Training	Bench Press		
CrossFit Session	Yoga Mat		

9. Premium Members

- **Purpose:**

Display premium members.

- **Screenshot:**

Results	Explain	Describe	Saved SQL	History
MEMBER_ID	EXTRA_FEATURES	PERSONAL_TRAINER		
2	Diet Plan + Personal Trainer	Trainer 1		
4	VIP Access	Trainer 2		

10. Feedback Records

- **Purpose:**

Display feedback.

- **Screenshot:**

Results	Explain	Describe	Saved SQL	History
FEEDBACK_ID	MEMBER_ID	COMMENTS	FEEDBACK_DATE	
1	1	Excellent Trainer	05-JUN-26	
2	2	Very Good Session	05-JUN-26	
3	3	Equipment Needs Upgrade	06-JUN-26	



11. View

- A SQL view was created.

➤ **MEMBER_VIEW**

- **Purpose:**

Shows member information with membership plan.

- **Includes:**

1. Member ID
2. Name
3. Email
4. Plan Name
5. Fee

- **Screenshot:**

Results	Explain	Describe	Saved SQL	History
MEMBER_ID	NAME	EMAIL	PLAN_NAME	FEE
1	Saad Malik	saad@gmail.com	Basic	3000
2	Hamza Ali	hamza@gmail.com	Premium	25000
3	Talha Khan	talha@gmail.com	Silver	8000
4	Zain Abbas	zain@gmail.com	VIP	40000
5	Omer Farooq	omer@gmail.com	Gold	15000



12. Trigger

- A trigger was created.

➤ CHECK_INVOICE_AMOUNT

- **Purpose:**
 - a) Prevent negative invoice amount.
 - b) Trigger checks invoice before insert/update.
 - c) If amount is negative:
 - d) Oracle raises error.
- **SS of test with negative value.**
- **Screenshot:**

Results Explain Describe Saved SQL History

```
ORA-20001: Amount cannot be negative
ORA-06512: at "SYSTEM.CHECK_INVOICE_AMOUNT", line 3
ORA-04088: error during execution of trigger 'SYSTEM.CHECK_INVOICE_AMOUNT'
1. INSERT INTO INVOICE
2. VALUES
3. (999,101,SYSDATE,-500);
```

Output of the Project

- The project **successfully**:
 - a) created all required tables
 - b) established relationships
 - c) inserted records
 - d) executed business queries
 - e) performed joins
 - f) used GROUP BY
 - g) used HAVING
 - h) executed subqueries
 - i) created view
 - j) implemented trigger
 - k) displayed outputs successfully



Conclusion

- The Gym Management System project was completed successfully using Oracle 10g Express Edition.
- The database stores and manages gym information efficiently.
- **Important achievements:**
 - a) relational schema created successfully
 - b) all relationships implemented
 - c) data inserted correctly
 - d) SQL queries executed successfully
 - e) view created
 - f) trigger validated successfully
- **This project improved understanding of:**
 - a) relational database design
 - b) SQL commands
 - c) Normalization
 - d) Joins
 - e) Constraints
 - f) Oracle database management
 - g) practical DBMS implementation
- Overall, the project achieved all required objectives successfully.

===== End of Report =====